



Using FEMTIC on NCI

Geophysical Research Data Processing and Modelling for 2030
Computation workshop

National Computational Infrastructure (NCI) Canberra, Australia

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CSIRO Mineral Resources

22/11/2023



I would like to begin by acknowledging the Traditional Owners of the land that we're meeting on today, and pay my respect to their Elders past and present.



Overview

FEMTIC

- Overview of the code
- Workflow to prepare and run inversions using *femticpy*:
 - MT data preparation
 - Mesh generation
 - Inversion on GADI
 - Results QC and visualization (Paraview)
- Synthetic example and jupyter notebook
- Test with an Australian dataset

Overview of the FEMTIC code

Geophysical Journal International



Geophys. J. Int. (2015) **202**, 828–849

doi: 10.1093/gji/ggv186

GJI Geomagnetism, rock magnetism and palaeomagnetism

3-D inversion of magnetotelluric data using unstructured tetrahedral elements: applicability to data affected by topography

Yoshiya Usui^{1,2}

Geophys. J. Int. (2017) **208**, 1359–1372

¹Department of Earth and Planetary Sciences, Advance Access publication 2016 December 9

²Volcanic Fluid Research Center, GJI Geomagnetism, rock magnetism and palaeomagnetism

Geophysical Journal International



doi: 10.1093/gji/ggv459

Three-dimensional resistivity structure of Asama Volcano revealed by data-space magnetotelluric inversion using unstructured tetrahedral elements

Yoshiya Usui,^{1,2} Yasuo Ogawa^{3,4}

Takao Koyama,⁵ Yu Takafumi Kasaya,^{3,4} and Hisanori Iwamoto⁴

¹Department of Earth and Planetary Sciences, Advance Access publication 2018 May 05

²Volcanic Fluid Research Center, GJI Marine geosciences and applied geophysics

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Geophysical Journal International



doi: 10.1093/gji/ggy171

Marine magnetotelluric inversion with an unstructured tetrahedral mesh

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FEMTIC is a 3-D magnetotelluric inversion code. FEMTIC is made by object-oriented programming with C++. FEMTIC is applicable to land magnetotelluric survey data as well as ocean bottom magnetotelluric survey data.

<https://github.com/yoshiya-usui/femtic>

sites.google.com/view/yoshiyausui/femtic



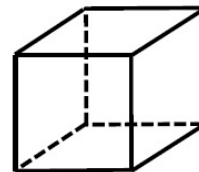
Functional overview

FEMTIC gives a three-dimensional electrical resistivity structure from the response functions at observation points on the Earth's surface.

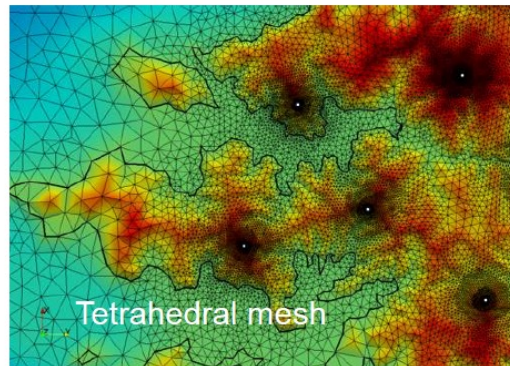
- Mesh type:
 - Tetrahedral mesh
 - Hexahedral brick mesh
 - Non-conforming deformed hexahedral mesh



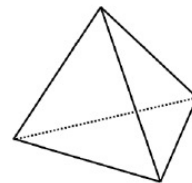
Brick hexahedral mesh



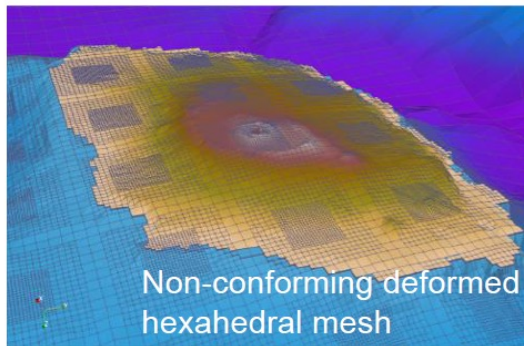
Brick hexahedral element



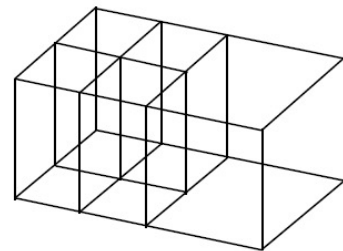
Tetrahedral mesh



Tetrahedral element



Non-conforming deformed hexahedral mesh



Non-conforming deformed hexahedral element

Functional overview

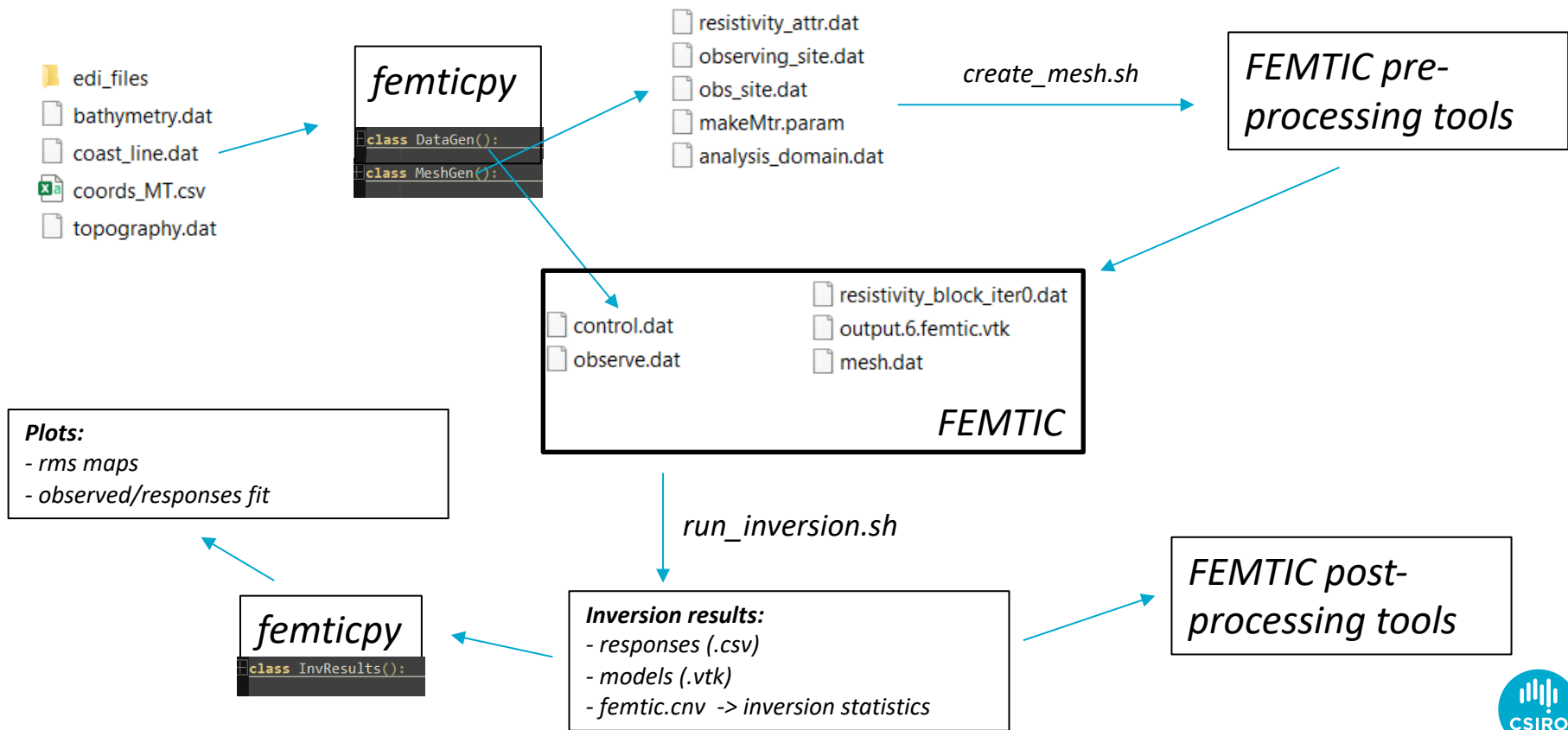
- FEMTIC gives a three-dimensional electrical resistivity structure from the response functions at observation points on the Earth's surface.
 - Mesh type: Tetrahedral mesh / Hexahedral brick mesh / Non-conforming deformed hexahedral mesh
 - Data type: Impedance tensor / Vertical magnetic transfer function / Inter-station horizontal magnetic transfer function / Phase tensor / Apparent resistivity & Phase
 - Inversion algorithm: Model-space Gauss-Newton method / Data-space Gauss-Newton method
 - Parallel computation: Multiple processes parallel computation with MPI / Multiple threads parallel computation with OpenMP / MPI & OpenMP hybrid parallel computation
 - Model parameter: Subsurface electrical resistivity / Distortion matrix of galvanic distortion
 - Regularization: L2 / L1 regularization

Workflow

- Workflow to prepare and run inversions using *femticpy*:
 - MT data preparation
 - Mesh generation on GADI
 - Inversion on GADI
 - Results QC and visualization (Paraview)

```
class DataGen():  
  
class MeshGen():  
  
class InvResults():
```

Workflow



Workflow: MT data preparation with *femticpy*

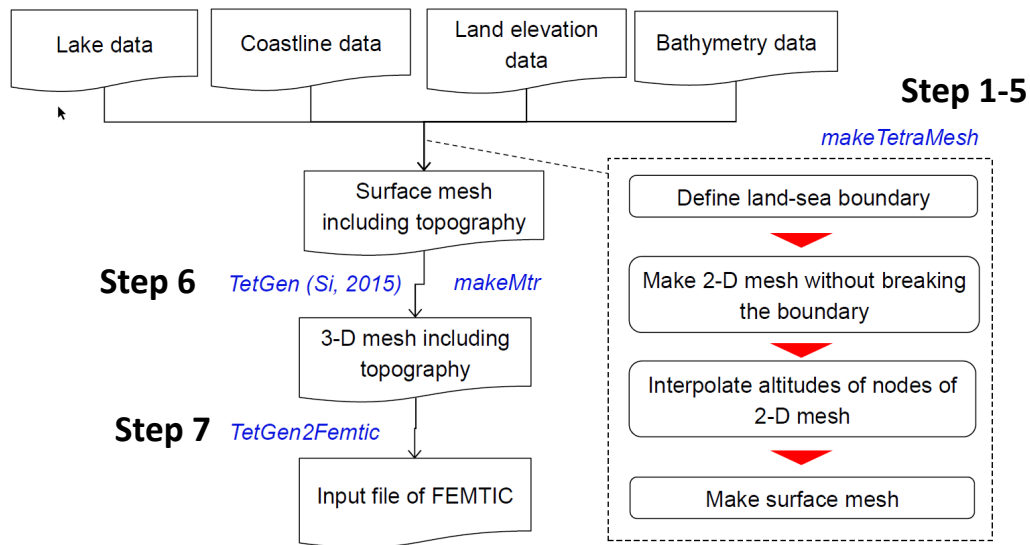
- EDI files → observe.dat
(only for impedance tensor)
- Set-up error floors
- Create Control file: Define inversion parameters

```
NUM_THREADS
1
MESH_TYPE
1
OUTPUT_PARAM_VTK
3
0 4 5
RESISTIVITY_BOUNDS
1
INV_METHOD
1
BOTTOM_RESISTIVITY
100.0
BOTTOM_ROUGHNING_FACTOR
1.0
DISTORTION
0
OUTPUT_OPTION
0 0
TRADE_OFF_PARAM
1.0
ITERATION
0 5
RETRIAL
2
CONVERGE
1.0
STEP_LENGTH
0.5 0.1 1.0
3
0.5 1.25
END
```

```
def write_inversion_control(self,
                            INV_METHOD = 1,
                            NUM_THREADS = 1,
                            DISTORTION = 0,
                            TRADE_OFF_PARAM = [3],
                            ITERATION = 10,
                            ):
    file_loc = os.path.join(self.outdir, 'control.dat')
    file = open(file_loc, 'w')
    file.write('NUM_THREADS\n')
    file.write('%d\n' % NUM_THREADS)
    file.write('MESH_TYPE\n')
    file.write('1\n')
    file.write('OUTPUT_PARAM_VTK\n')
    file.write('2\n')
    file.write('0 4\n')
    file.write('OFFILE_TYPE\n')
    file.write('0\n')
    file.write('DISTORTION\n')
    file.write('%d\n' % DISTORTION)
    file.write('INV_METHOD\n')
    file.write('%d\n' % INV_METHOD)
    file.write('BOTTOM_RESISTIVITY\n')
    file.write('100.0\n')
    file.write('BOTTOM_ROUGHNING_FACTOR\n')
    file.write('1.0\n')
    file.write('OUTPUT_OPTION\n')
    file.write('0 0\n')
    file.write('TRADE_OFF_PARAM\n')
    for i in range(len(TRADE_OFF_PARAM)):
        file.write('%0.2f ' % TRADE_OFF_PARAM[i])
    file.write('\n')
    file.write('ITERATION\n')
    file.write('0 %d\n' % ITERATION)
    file.write('RETRIAL\n')
```

Workflow: Mesh generation

How to make 3-D mesh including topography



edi_files
bathymetry.dat
coast_line.dat
coords_MT.csv
topography.dat

femticpy
class MeshGen():

Input files for
meshGen.sh:

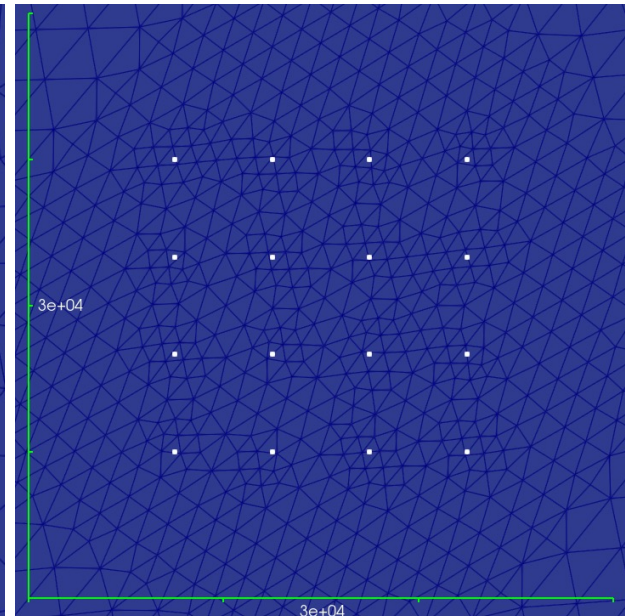
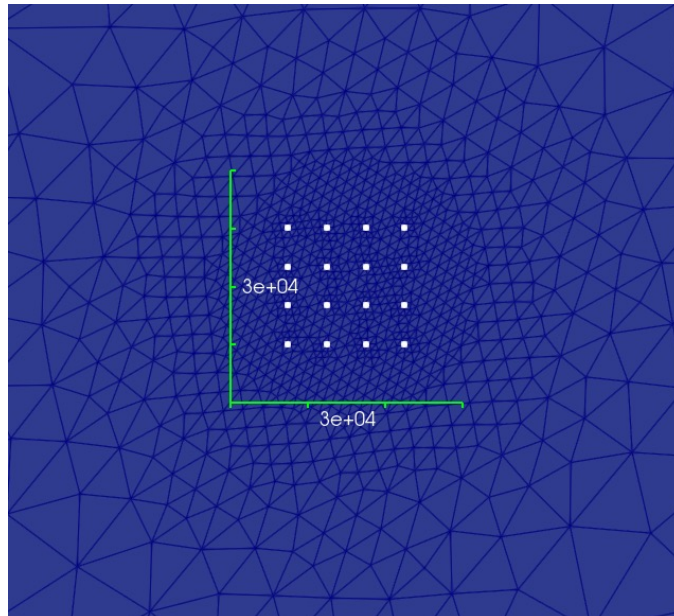
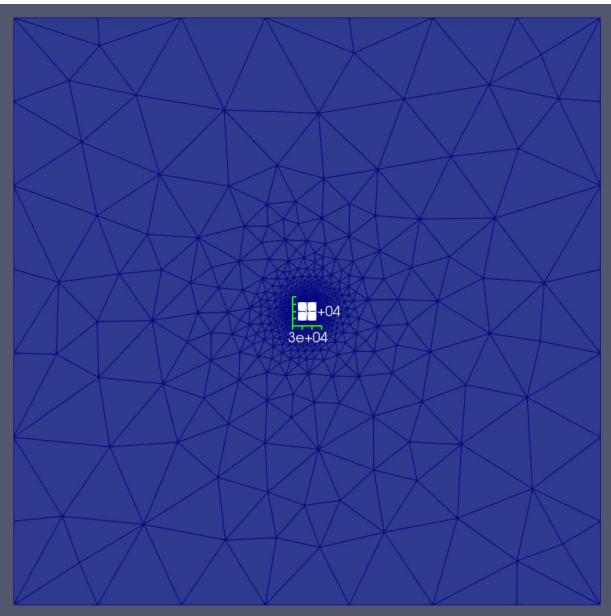
Script that sequentially runs the different pre-processing routines required to create the mesh.

meshGen.sh is ran as a job on Gadi.

Workflow: Mesh generation

edi_files
bathymetry.dat
coast_line.dat
coords_MT.csv
topography.dat

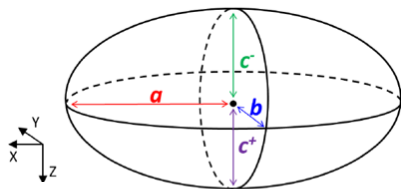
femticpy



Workflow: Mesh generation

edi_files
 bathymetry.dat
 coast_line.dat
 coords_MT.csv
 topography.dat

femticpy



a len f_h f_v^+ f_v^-

a : Length along x axis (km)

len : Edge length within the ellipsoid (km)

f_h : Oblateness on the X-Y plane

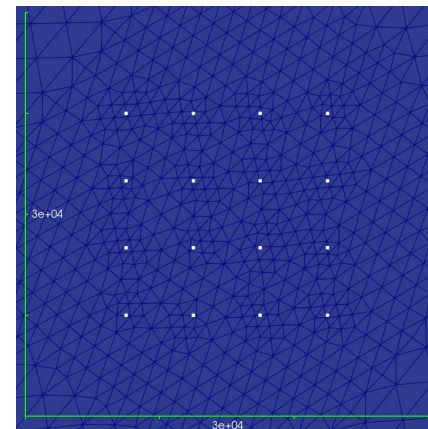
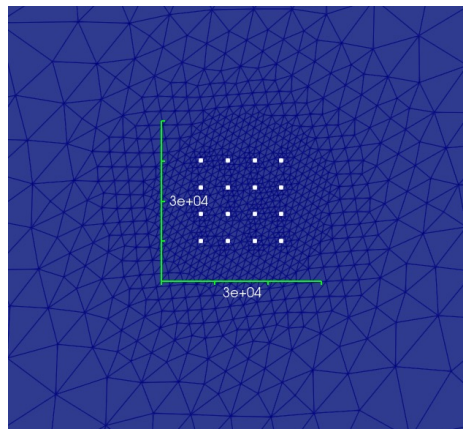
f_v^+ : Oblateness on the Z-X plane (Upper side)

f_v^- : Oblateness on the Z-X plane (Lower side)

$$\frac{b}{a} = 1 - f_h$$

$$\frac{c^+}{a} = 1 - f_v^+$$

$$\frac{c^-}{a} = 1 - f_v^-$$



Step 1-5

```

mesh01.center = [0.0, 0.0, 0.0]
mesh01.rotation = 0
mesh01.ellipsoids_control = [3,
                             [15, 2, 0.0, 0.0, 0.0],
                             [30, 5, 0.0, 0.0, 0.0],
                             [200.0, 100.0, 0.0, 0.0, 0.0]]
  
```

```

mesh01.ellipsoids_observing_sites = [1,
                                     1., 1.]
  
```

Step 6

```

mesh01.ellipsoids_obs_sites = [1,
                               [1., 0.5, 0.3]]

# and in the full mesh
mesh01.ellipsoids_mtr = [4,
                        [10, 2, 0.0, 0.0, 0.0],
                        [30, 5, 0.0, 0.0, 0.0],
                        [100.0, 100.0, 0.0, 0.0, 0.0],
                        [300, 100.0, 0.0, 0.0, 0.0]]
  
```

Step 7

```

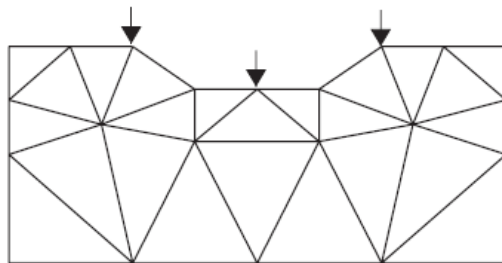
mesh01.ellipsoids_resistivity_attr = [3,
                                     [15.0, 2.0, 0.0, 0.0],
                                     [30, 5.0, 0.0, 0.0],
                                     [300.0, 300.0, 0.0, 0.0]]

mesh01.ellipsoids_resistivity_attr_sites = [1,
                                             [1.0, 1.0]]
  
```

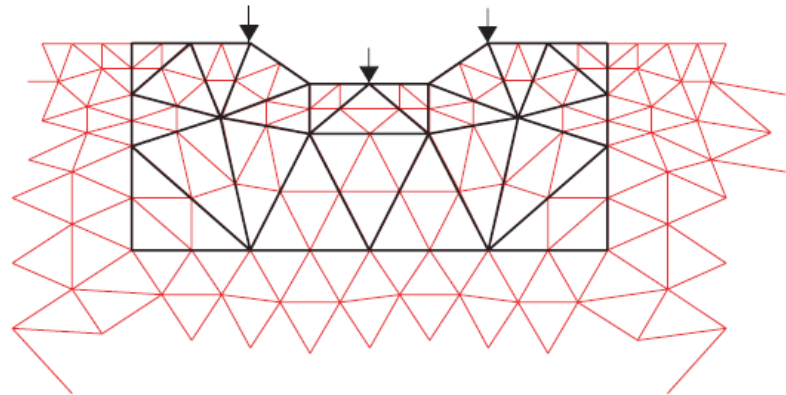
Workflow: Mesh generation

edi_files
bathymetry.dat
coast_line.dat
coords_MT.csv
topography.dat

femticpy



Günther, T., Rücker, C. & Spitzer, K., 2006. Three-dimensional modelling and inversion of dc resistivity data incorporating topography – II. Inversion, *Geophys. J. Int.*, **166**(2), 506–517.



Step 1-5

```
mesh01.center = [0.0, 0.0, 0.0]
mesh01.rotation = 0
mesh01.ellipsoids_control = [3,
                             [15, 2, 0.0, 0.0, 0.0],
                             [30, 5, 0.0, 0.0, 0.0],
                             [200.0, 100.0, 0.0, 0.0, 0.0]]
```

```
mesh01.ellipsoids_observing_sites = [1,
                                     1., 1.]
```

Step 6

```
mesh01.ellipsoids_obs_sites = [1,
                               [1., 0.5, 0.3]]

# and in the full mesh
mesh01.ellipsoids_mtr = [4,
                        [10, 2, 0.0, 0.0, 0.0],
                        [30, 5, 0.0, 0.0, 0.0],
                        [100.0, 100.0, 0.0, 0.0, 0.0],
                        [300, 100.0, 0.0, 0.0, 0.0]]
```

Step 7

```
mesh01.ellipsoids_resistivity_attr = [3,
                                     [15.0, 2.0, 0.0, 0.0],
                                     [30, 5.0, 0.0, 0.0],
                                     [300.0, 300.0, 0.0, 0.0]]

mesh01.ellipsoids_resistivity_attr_sites = [1,
                                             [1.0, 1.0]]
```

Workflow: Generate the mesh on GADI

```
[hs3588@gadi-login-03 hs3588]$ qsub create_mesh.sh
```

Output:

- inversion files:
 - resistivity_block_iter0.dat
 - mesh.dat
- Visualization files (Paraview):
 - output.6.femtic.vtk
 - triangles_with_height.vtk

```
#!/bin/bash
```

```
project=$1
cd $project
cp input_data/{topography.dat,bathymetry.dat,coast_line.dat} ./meshGen/files/
cd meshGen
```

```
#-----"
#----  STEP 1  ----"
#-----"
```

```
echo "STEP 1... "
cp files/{control.dat,coast_line.dat,analysis_domain.dat,observing_site.dat} .
makeTetraMesh -stp 1
echo "   done!"
sleep 1
```

```
#-----"
#----  STEP 2  ----"
#-----"
```

```
echo "STEP 2... "
makeTetraMesh -stp 2
```

```
[hs3588@gadi-login-03 meshGen]$ head resistivity_block_iter0.dat
28190    3933
  0         0
  1         0
  2    1701
  3         0
  4    3559
  5    1128
  6         0
  7         0
  8         0
[hs3588@gadi-login-03 meshGen]$
```

resistivity_block_iter[Iter#].dat

Number of elements (N_e)

Number of parameter cells (N_p)

Workflow: Inversion on GADI

```
[hs3588@gadi-login-03 hs3588]$ qsub run_inversion.sh
```

Output, for each iteration:

- responses (.csv)
- models (.vtk)
- obs_loc.vtk
- femtic.cnv -> inversion statistics

```
#!/bin/bash
#PBS -P nm05
#PBS -q normal
#PBS -l ncpus=8,walltime=0:30:00,mem=20GB,jobfs=1GB
#PBS -l storage=scratch/nm05+gdata/up99+gdata/nm05

module use /g/data/up99/modulefiles
module load femtic/4.1

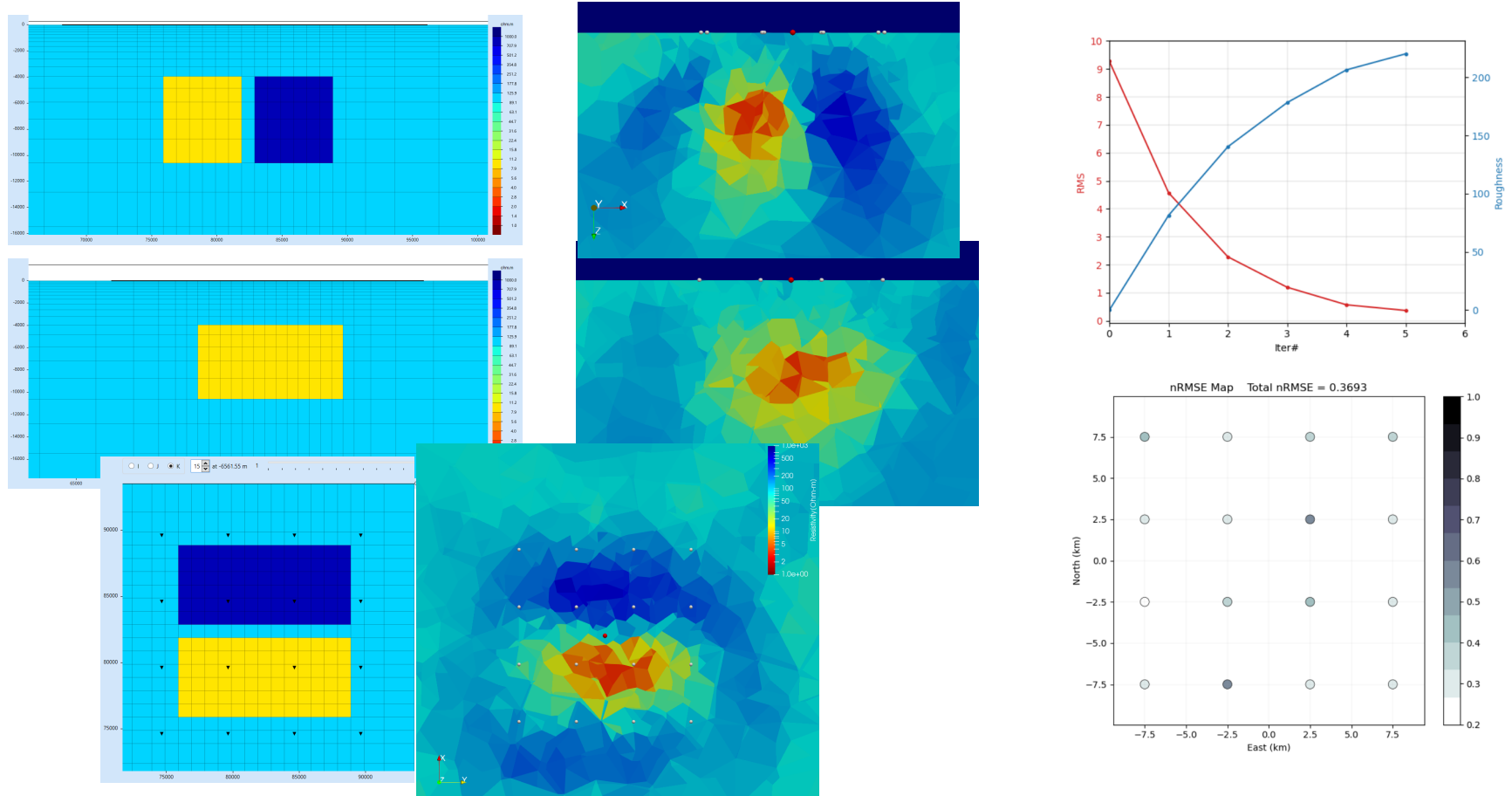
project=synthetic

cd ${PBS_O_WORKDIR}

cd $project/inversion

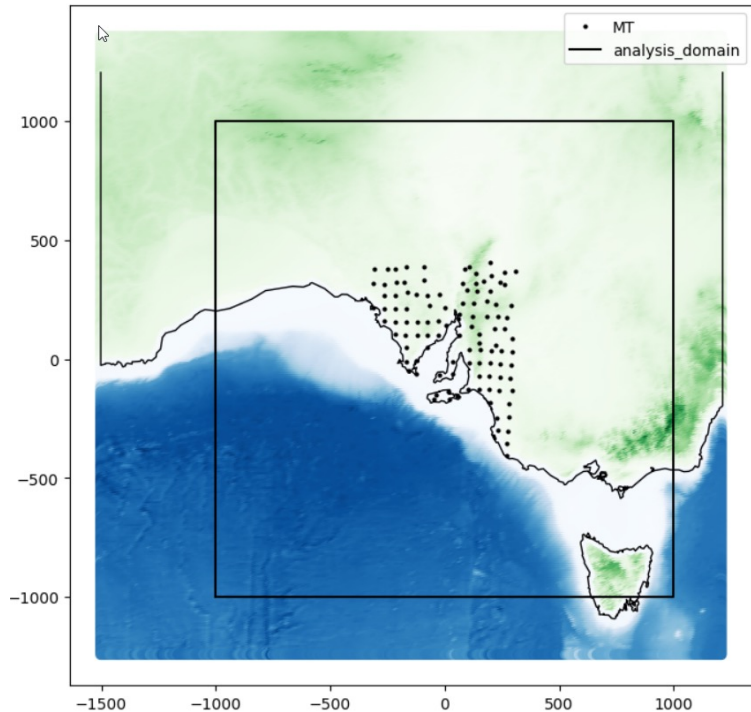
mpirun -np 8 femtic
```

Workflow: Results QC and visualization (Paraview)



Synthetic example and jupyter notebook

Test with an Australian dataset



```
inversion01.error_floor_Z = [0.1, 0.05, 0.05, 0.1]
```

109 MT sites

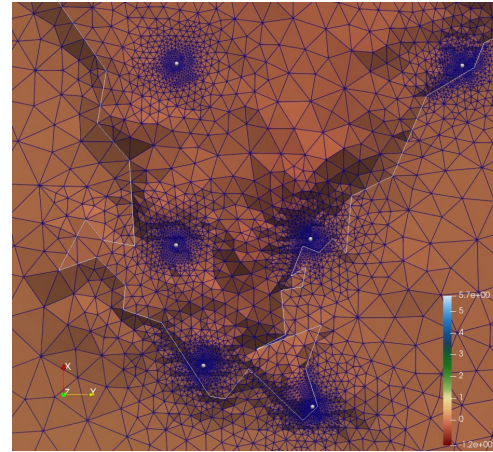
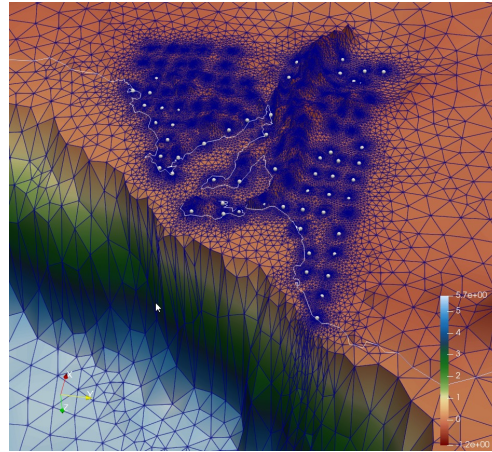
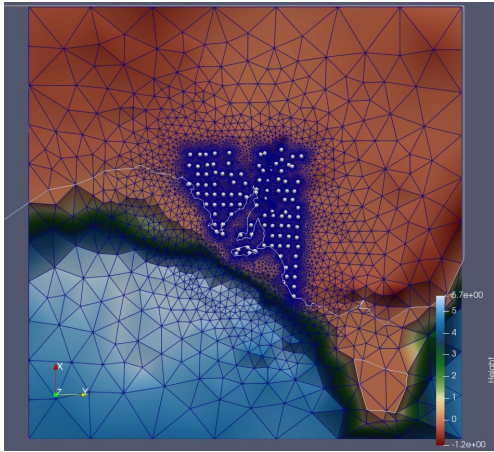
19 freqs / site, in between 0.03125 Hz and 0.00006 Hz

```
#SBATCH --ntasks-per-node=16
#SBATCH --nodes=2
#SBATCH --mem=900g
```

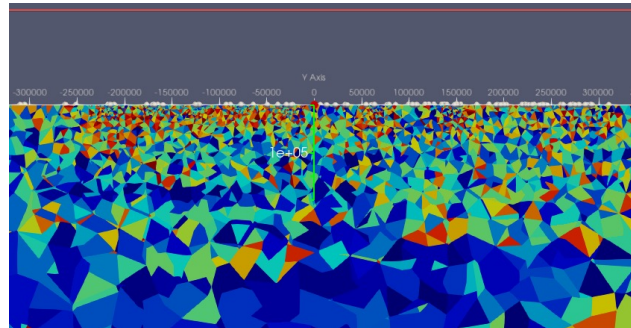
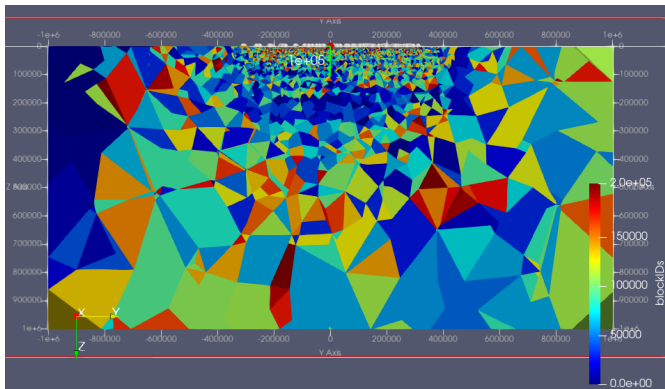
```
# Application specific commands:
module load intel-fc
module load intel-cc
module load intel-mkl/2020.1.217
module load openmpi/4.1.1-ofed51-intel20
```

```
mpirun -n 32 ../femtic
```

Test with an Australian dataset

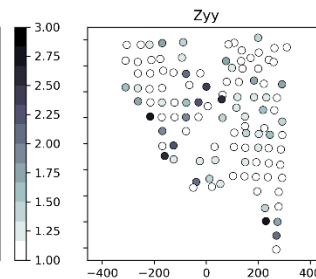
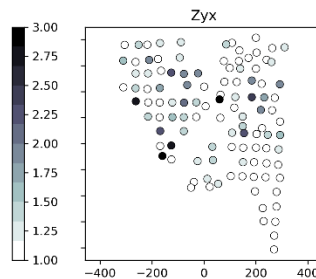
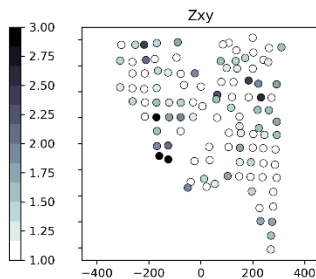
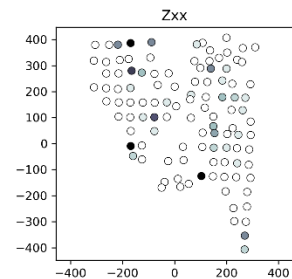
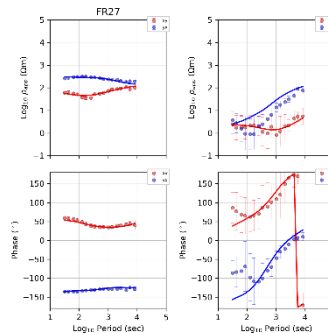
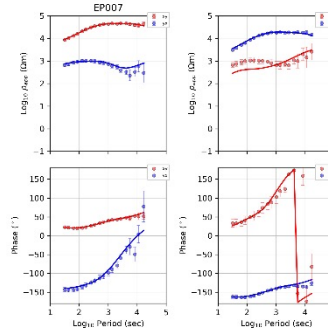
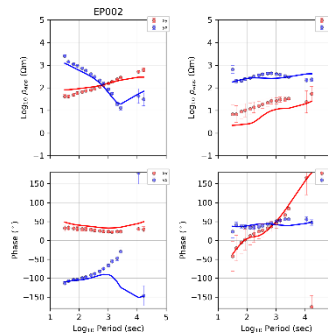
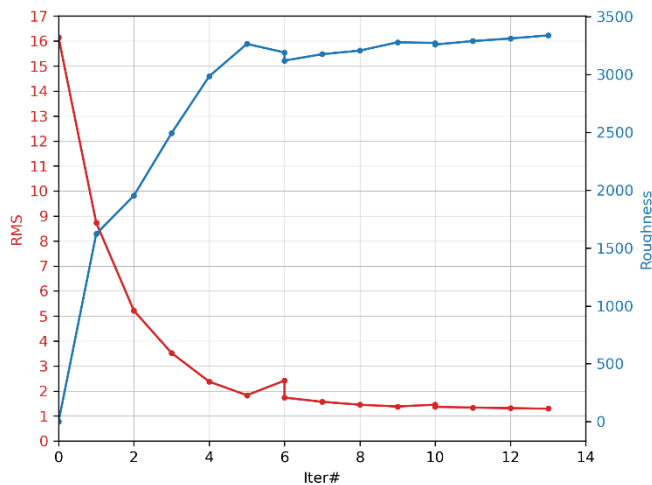
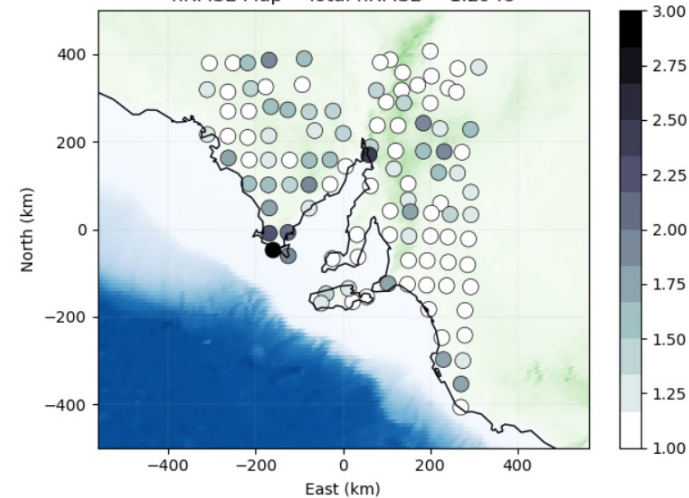


200k Cells



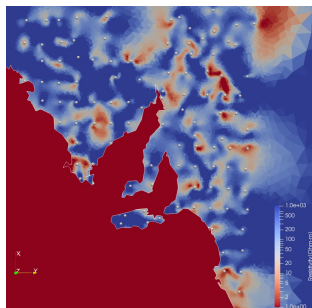
Test with an Australian dataset

nRMSE Map Total nRMSE = 1.2948

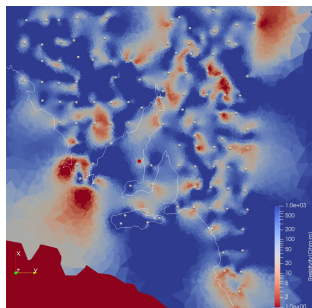


Test with an Australian dataset

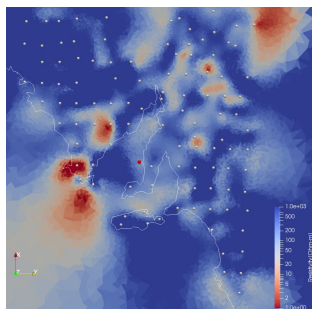
0 km



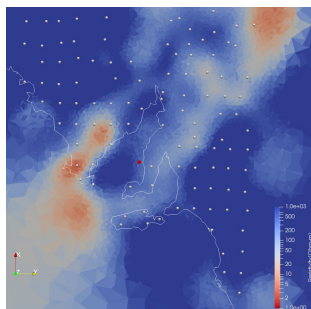
5 km



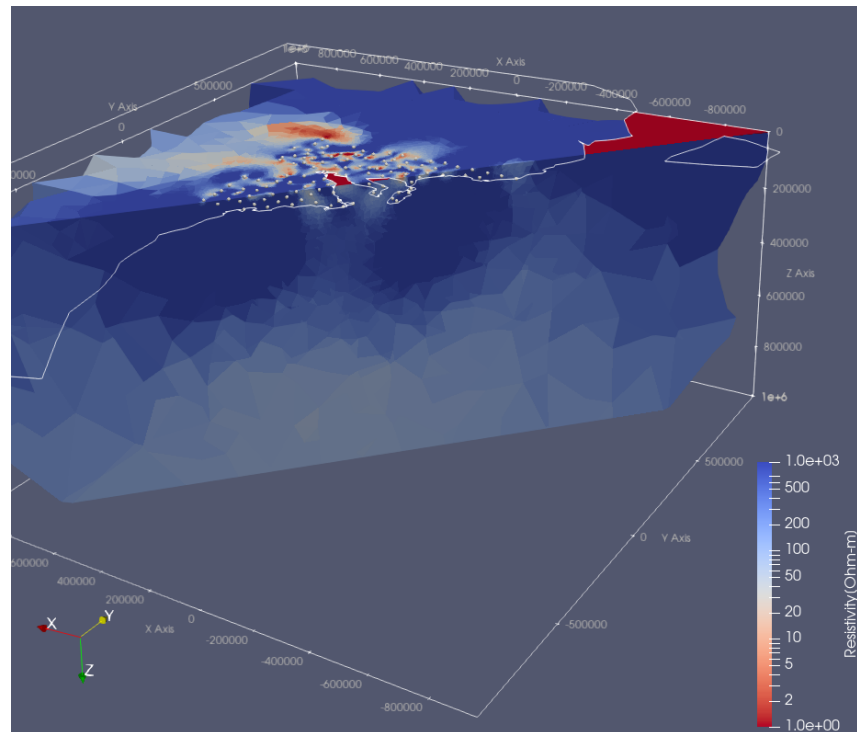
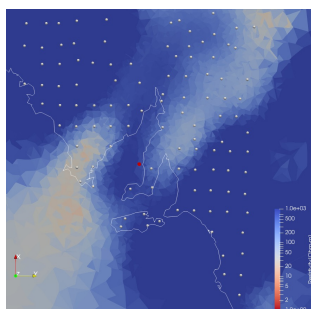
20 km



50 km



100 km





Thank you

Hoël Seillé - Research Scientist
CSIRO Mineral Resources

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